

A Comparative Study of Supervised Machine Learning Models for Stroke Risk Prediction

Vanshika Tyagi
Artificial Intelligence
Indira Gandhi Delhi Technical
University for Women
Delhi, India
vansutyagi5517@gmail.com

Shubhangi Jha
Artificial Intelligence
Indira Gandhi Delhi Technical
University for Women
Delhi, India
shubhangijha4444@gmail.com

Abstract— Around the world, stroke is one of the primary contributors to mortality and impairment, thus accurate and early stroke detection methods are necessary to improve patient outcomes. In this study, we present a machine learning-based approach to predict stroke risk in patients using clinical and demographic data. This study employs models built upon supervised learning algorithms like Logistic Regression, Random Forest and Support Vector Machines to analyse parameters like age, gender, hypertension, cardiac disease, body mass index, glucose levels and smoking patterns. The dataset used for training and evaluation was derived from publicly available healthcare repositories, and missing values and class imbalance was handled by data pre-processing. The performance of the model was assessed using a range of parameters like accuracy, precision, recall, F1-score and the ROC-AUC score. Our analysis indicates that machine learning models, particularly ensemble methods, may accurately detect patterns linked with stroke risk and provide reliable predictions. This study demonstrates the prospects of AI-powered solutions in supporting healthcare professionals with early stroke detection, allowing for prompt medical intervention and improved patient care.

I. INTRODUCTION

Stroke is one of the primary causes of mortality and long-term neurological impairments. According to estimates by the World Health Organization (WHO), roughly 15 million individuals are affected by stroke annually. Out of these around 5 million result in fatalities, leaving another 5 million with permanent impairments. A stroke occurs upon interruption of supply of blood to a region of the brain, leading to oxygen and nutrient deprivation in brain tissue, which in turn leads to cell death. If not treated, a stroke may lead to severe complications, often resulting in sudden loss of brain functions or even death. In order to improve early detection and possibly save lives, this study examines the application of machine learning models for prediction of stroke risk using publicly available datasets. Machine learning has shown immense potential as an effective stroke diagnosis tool in recent years. Doctors can proactively identify high-risk individuals and take steps to avoid before a stroke takes place by integrating these technologies into clinical settings.

The structure of this paper is as follows: Section 2 presenting the literature review, summarizing prior research relevant to the study. Section 3 outlines the methodology—details about the dataset, data preprocessing and the machine learning models implemented. Section 4 discusses the experimental outcomes obtained from the applied models. The study is concluded with its potential direction for future research under Section 5.

II. LITERATURE REVIEW

Since prompt diagnosis and intervention are so important, utilizing supervised machine learning approaches

in early stroke risk prediction has gained significant interest. Several studies have investigated numerous models, datasets, and preprocessing techniques to enhance clinical applicability and prediction accuracy.

On structured and imaging datasets, Al-Mekhlafi et al.[1] used both deep learning and conventional machine learning classifiers, like Support Vector Machines. They stated in their study that ensemble techniques like Random Forests produced better results and underlined the significance of data preprocessing, namely using KNN to fill in missing BMI values. Similarly, Dritsas and Trigka[2] demonstrated the effectiveness of model ensembling by using a Naive Bayes classifier and proposing a stacking model that outperformed individual learners by a large margin.

In the field of medical imaging, by applying logistic regression to MRI datasets, Liu et al.[3] demonstrated precise identification of early stroke-warning symptoms by machine learning models utilizing radiomic characteristics and lesion segmentation. Sheth et al.[4] analysed various machine learning methods applied to acute stroke imaging, highlighting the models' versatility as diagnostic tools.

Adam et al.[5] used a Decision Tree classifier on 400 ischemic stroke patients' data. Their study achieved strong classification performance, validating the efficacy of simple interpretable models in clinical contexts. Rahman et al.[6] used a Kaggle dataset to test Random Forest and XGBoost. The Random Forest model demonstrated its potential for public health applications by yielding the highest accuracy.

Latha et al.[7] worked on B-mode ultrasound imaging and applied CART Decision Trees to detect carotid artery plaque, achieving classification accuracies of over 90%. Tazin et al.[8] used a dual-input residual convolutional model (BCNN-ResNet) on Doppler ultrasound images for carotid plaque detection, achieving an AUC of 0.92.

Roohi et al.[9] simulated data using CST Studio to develop a delay-multiply-and-sum algorithm that achieved a classification accuracy of 91.5%. Alon and Dehkharghani[10] implemented a finite element model (FEM) with an 8-head antenna array, achieving 94.6% accuracy in stroke classification based on electromagnetic simulations.

Mainali et al.[11] reviewed a range of ML techniques including RF, CNNs, and SVMs. They found ensemble models performed best overall. Dev et al.[12] analyzed EHRs using linear vector quantization and neural networks. Neural networks showed the highest performance, with accuracy reaching 78%.

Hassan et al.[13] applied both logistic regression and advanced gradient descent models (LR-AGD), achieving 96.13% accuracy on an imbalanced dataset. Ozaltin et

al.[14] introduced OzNet, a custom CNN architecture trained on CT images. It outperformed standard CNNs with an accuracy of 87.47%. Finally, Park et al.[15] built a signal-acquisition model using wrist-worn sensors. Without feature selection, their models still achieved AUC scores between 0.81 and 0.91, suggesting high promise in wearable-tech-based stroke prediction.

III. METHODOLOGY

A. Dataset

This study utilised publicly available dataset Healthcare Stroke Dataset on Kaggle. It contains medical and demographic information for 5,110 individuals, including attributes commonly associated with stroke risk.

The dataset includes 12 attributes, out of which 11 are considered relevant for the prediction task: gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status, and the target variable stroke (with 1= stroke and 0=no stroke). The complete list of attributes along with their respective data types is presented in Table 1.

TABLE I. Dataset's description attributes and their data types

ATTRIBUTE	DATATYPE
id	int64
gender	object
age	float64
hypertension	int64
heart_disease	int64
ever_married	object
work_type	object
residence_type	object
avg_glucose_level	float64
bmi	float64
smoking_status	object
stroke	int64

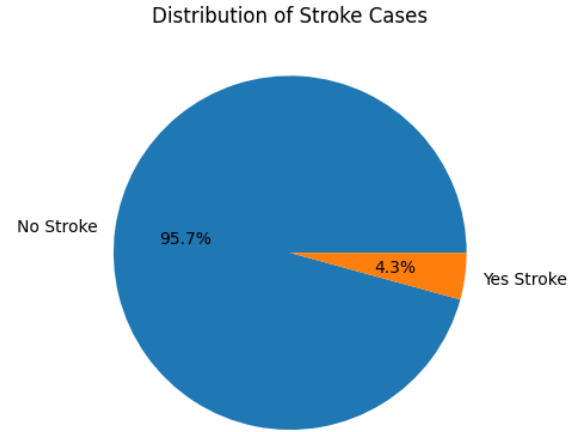
As shown in Table 1, the dataset combines both quantitative features (e.g., age, glucose level) and qualitative features (e.g., gender, type of employment). These features represent both clinical and socio-economic factors relevant for stroke prediction.

B. Data Pre-Processing

Before model training, the dataset underwent preprocessing to improve data quality as well as to ensure consistency. The bmi column contained missing values, which were handled by dropping the affected rows due to their relatively small proportion.

The qualitative parameters like gender, marital status, type of employment, type of residence, and smoking behaviour were encoded using label encoding or one-hot encoding as appropriate. Continuous variables like age and avg_glucose_level were normalized to achieve consistent scaling of features. The preprocessed dataset was subsequently divided

into training as well as testing sets for evaluating model performance.



- Pie chart showing the proportion of patients with and without a history of stroke

Figure 1 highlights the **imbalance in class distribution**, with stroke cases accounting for only **4.3%** of the dataset. This significant disparity between the 'stroke' and 'no stroke' classes can affect model performance, particularly for algorithms sensitive to class imbalance. Addressing this imbalance through appropriate evaluation metrics or sampling techniques is essential for developing reliable predictive models.

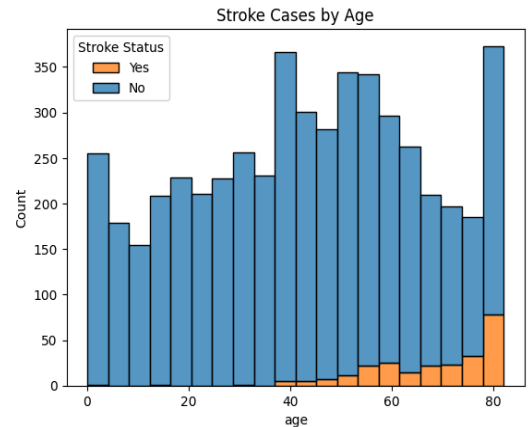


Fig. 1. Stacked histogram comparing stroke prevalence across different age groups

Figure 2 indicates a **clear spike in stroke occurrence with age**. Stroke cases are minimal among younger individuals but rise steadily in older age brackets, particularly beyond the age of 60. This trend reinforces the strong association between **aging and stroke risk**, making age a critical feature in prediction models.

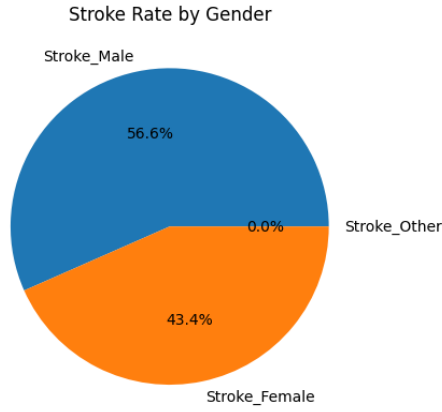


Fig. 2. Distribution of stroke cases by gender

As suggested by **Figure 3**, female patients account for a slightly higher proportion of stroke cases compared to males. This observation may indicate a possible gender-related trend in stroke occurrence, warranting further investigation into how biological and lifestyle factors associated with gender could influence stroke risk.

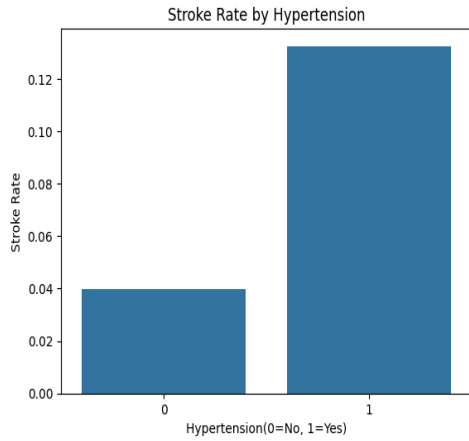


Fig. 3. Bar chart showing stroke rates for patients with and without hypertension

Figure 4 demonstrates a clear disparity in stroke rates based on hypertension status. Individuals with hypertension (1) appear to have a noticeably higher stroke rate, reinforcing its role as a critical risk factor in model training and clinical screening.

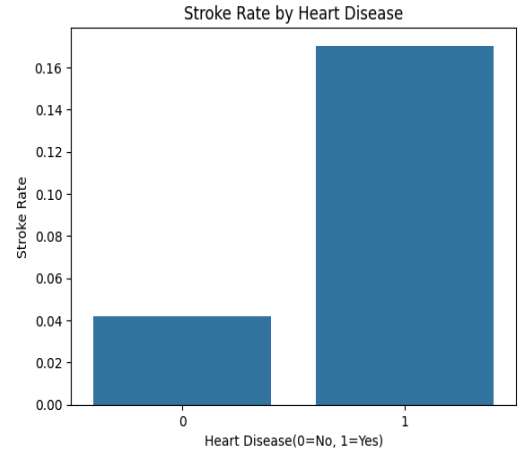


Fig. 4. Stroke rate comparison between patients with and without heart disease

According to **Figure 5**, individuals with underlying cardiac conditions(1) exhibit a notably **higher stroke rate** than those without (0). This observation aligns with medical literature, where pre-existing cardiovascular conditions are considered a **significant contributor to stroke risk**. The difference supports the inclusion of heart disease as a critical variable influencing the model.

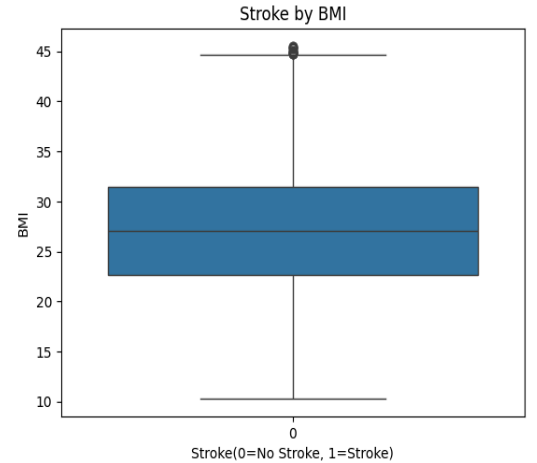


Fig. 5. Box plot representing the BMI distribution stratified by stroke status

As depicted in **Figure 6**, there is **some overlap** in BMI distributions for stroke and non-stroke patients. However, stroke cases show a **slightly elevated median BMI**, suggesting that higher body mass may contribute marginally to stroke risk. While BMI alone may not be a decisive factor, it could still provide **supporting predictive value** when combined with other health indicators.

C. Models Implemented

In this study, two supervised learning models—Random Forest Classifier and Logistic Regression— were applied to predict the probability of a patient experiencing a stroke, on the basis of numerous health and demographic parameters from a medical dataset.

Random Forest constructs multiple decision trees, training each tree on a randomly drawn subset of the dataset and features, enhancing the model's generalization capability. The output is determined by majority voting of outputs from all the trees, enhancing model reliability and reducing the risk of overfitting. This model is very effective for handling imbalanced datasets, which is common in healthcare-related data. Post model-training, the performance was assessed using evaluation parameters such as accuracy, confusion matrix, and classification report (precision, recall, F1-score).

The second model implemented was Logistic Regression, a commonly applied linear algorithm for binary decision making. It works by estimating the probability that a data point belongs to one of two classes—stroke or no stroke—using the logistic (sigmoid) function to produce output probabilities. Logistic Regression is easy to use and performs well with linearly separable data. Before training, StandardScaler was used to normalize the features by centering them around mean and further scaling to have unit variance—which is important since Logistic Regression is sensitive to feature scales. Once trained, the performance of the model was evaluated by comparing its outcomes to the actual outcomes using classification metrics to evaluate its effectiveness.

Both models contributed valuable insights. Logistic regression helps in performing the function related to linearly separable data and the Random Forest model provides ease in handling complex data. When both of the models are used they help in identifying the stroke risk factors.

IV. RESULTS

Results of the models after evaluating the dataset is summarised in Table 2:

TABLE II. Accuracy scores of the models used for stroke prediction

MODEL	ACCURACY(%)
Logistic Regression	95.012
Random Forest Classifier	97.377
K-Nearest Neighbour Classifier	97.069
Decision Tree Classifier	95.167

Table 2 presents the accuracy scores obtained by the machine learning models applied in this study. Among the four classifiers, the Random Forest classifier has the accuracy of 97.38%, showing how reliable it is at identifying non-linear patterns and feature interactions.

A comparable accuracy of 97.07% was also achieved by the K-Nearest Neighbors (KNN) classifier, whereas 95.01% and 95.17% were attained by the Logistic Regression and Decision Tree models.

These results show that using these models we can predict the result of stroke problems.

V. CONCLUSION AND FUTURE SCOPE

Using structured health data, this study explored the potential of machine learning approaches for early stroke risk prediction. With the accuracy of 97.38%, the Random Forest classifier outperformed the remaining models used, closely followed by K-Nearest Neighbors and Logistic

Regression. The findings show that health and demographic factors can be analyzed and patterns suggestive of stroke risk can be found using both ensemble and linear models.

These results indicate that early intervention and clinical decision-making can be significantly enhanced by machine learning. The models' feature importance may further help in emphasizing important risk factors.

Future research can incorporate more clinical data like cholesterol levels, medication history, etc. to improve the predictive framework. The application of deep learning architectures is promising, especially when interacting with multi-modal data such as sensor and medical imaging data. Such models can be integrated into wearable health devices or telemedicine systems for stroke risk monitoring, facilitating real-world adoption. Comprehensive evaluation using bigger, more varied datasets will further enhance the clinical utility.

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